

A Project Report On :

**“ Developement of smart digital platform to
promote eco and cultural tourism in Jharkhand “**

Submitted by :

Sr.No.	Name
1	Nikita Shelar
2	Sneha Shelar
3	Supriya Dakhane
4	Dhanshri Bajare

Introduction

This project is developed as part of the **Java Full Stack Training Program conducted by Symbiosis University**. The training aims to provide practical knowledge of full stack development using modern technologies such as Java, Spring Boot, Angular, and MySQL.

As part of this program, the project titled **“Development of Smart Digital Platform to Promote Eco and Cultural Tourism in Jharkhand”** focuses on solving real-world problems using web-based solutions. The project integrates both frontend and backend technologies to build a complete application that provides tourism-related information, trip planning.

The main purpose of this project is to apply the concepts learned during training to develop a user-friendly and efficient platform. It also helps in understanding real-time application development, database management, API integration, and system design.

Through this project, students gain hands-on experience in building scalable web applications while contributing to promoting eco-friendly and cultural tourism.

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Abstract :

Tourism plays an important role in the economic development of any region. Jharkhand is a state rich in natural beauty, forests, waterfalls, wildlife, and diverse cultural heritage. However, due to the lack of proper digital platforms and awareness, many tourist destinations remain unexplored.

This project aims to develop a smart digital platform that promotes eco-tourism and cultural tourism in Jharkhand. The platform will provide comprehensive information about tourist places, cultural events, traditions, and eco-friendly travel options. It will also include features such as user registration, login, destination search, itinerary planning, and user feedback.

By integrating modern web technologies like Angular, Spring Boot, and MySQL, the system ensures a responsive and efficient user experience. This platform will help tourists make informed decisions and encourage sustainable tourism while supporting local communities.

SOFTWARE REQUIREMENT :

- Operating System: Windows 10/11 or Linux
- Programming Languages: Java, HTML, CSS, TypeScript
- Frontend: Angular
- Backend: Spring Boot (Java)
- Database: MySQL
- IDE/Tools: VS Code, IntelliJ IDEA/Eclipse
- Testing Tools: Postman

Problem Statement :

Jharkhand has immense potential in tourism due to its natural landscapes and rich cultural traditions. However, several problems limit its growth:

- Lack of centralized digital platform for tourism information
- Limited awareness about eco-tourism and cultural heritage
- Difficulty in trip planning due to scattered information
- No proper system for booking and itinerary management
- Poor promotion of local festivals, crafts, and traditions
- Limited support for local businesses and guides

Due to these challenges, tourists often prefer other destinations with better digital accessibility. Therefore, there is a need to develop a unified, smart, and user-friendly platform that can address these issues effectively.

Objective :

The main objectives of this project are:

- To design and develop a centralized digital platform for tourism in Jharkhand
- To provide detailed and accurate information about eco-tourism destinations
- To highlight cultural heritage such as festivals, traditions, and art forms
- To enable easy trip planning through itinerary generation
- To promote eco-friendly tourism practices among users
- To enhance user experience through an interactive interface
- To support local communities by increasing tourism engagement
- To implement a scalable and efficient system using modern technologies

Theory :

The project is based on the concept of Smart Tourism Systems, which use digital technologies to enhance tourist experience and improve tourism management.

Key Theoretical Concepts:

1. Smart Tourism

Smart tourism integrates information and communication technologies (ICT) to provide personalized and efficient services to tourists.

2. Eco-Tourism

Eco-tourism focuses on responsible travel to natural areas that conserve the environment and improve the well-being of local people.

3. Cultural Tourism

Cultural tourism involves exploring traditions, festivals, heritage sites, and local lifestyles.

4. Web-Based Application Architecture

The system follows a three-tier architecture:

- Presentation Layer (Frontend - Angular)
- Application Layer (Backend - Spring Boot)
- Data Layer (Database - MySQL)

5. Database Management System (DBMS)

Used to store and manage data such as users, locations, bookings, and reviews efficiently.

6. RESTful API

Used for communication between frontend and backend, ensuring scalability and flexibility.

Implementation Steps :

Step 1: Requirement Analysis

- Identify system users (Tourists, Admin)
- Define features like login, search , reviews
- Gather functional and non-functional requirements

Step 2: System Design

- Design system architecture
- Create UML diagrams (Use Case, Class Diagram)
- Design database schema and ER diagram

Step 3: Frontend Development (Angular)

- Develop user interface pages:
 - Home Page
 - Login/Register Page
 - Tourist Places Page
- Implement navigation and responsive design
- Use Angular services for API integration

Step 4: Backend Development (Spring Boot)

- Create REST APIs for:
 - User authentication
 - Place management
- Implement business logic
- Ensure security and validation

Step 5: Database Implementation (MySQL)

- Create tables:
 - Users
 - Tourist Places
 - Bookings
 - Reviews
- Establish relationships between tables

Step 6: Integration

- Connect Angular frontend with Spring Boot backend using APIs
- Test API responses and data flow

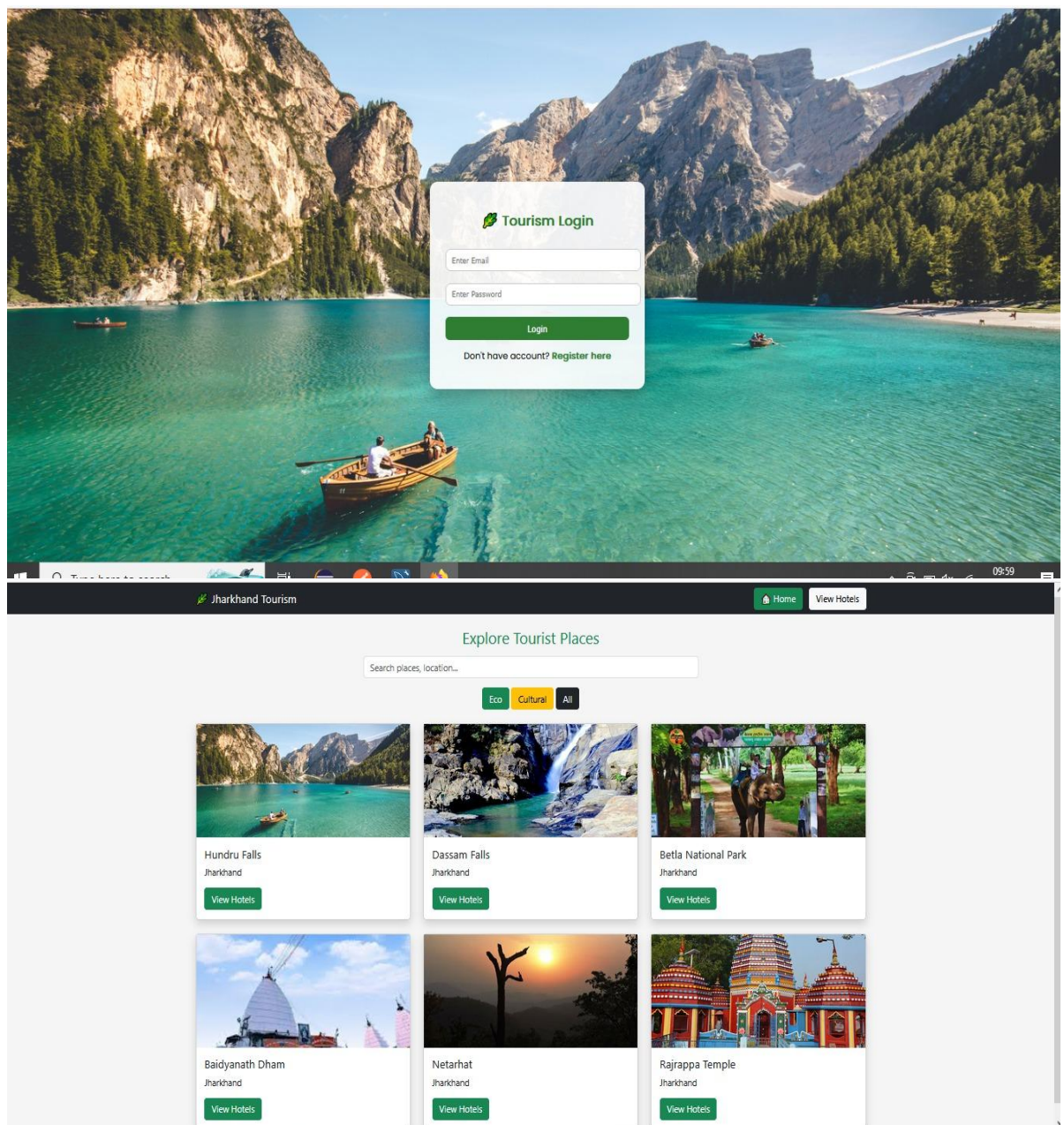
Step 7: Testing

- Perform unit testing and integration testing
- Validate user inputs and system outputs
- Fix bugs and improve performance

Outcome :

The project results in a fully functional web-based platform that:

- Provides detailed information about tourist destinations
- Enables easy navigation and search functionality
- Allows users to plan trips efficiently
- Supports booking and itinerary management
- Promotes eco-friendly and cultural tourism
- Enhances user engagement through reviews and ratings
- Improves visibility of Jharkhand tourism
- Supports local economy and small businesses



Conclusion :

The development of this smart digital platform successfully addresses the challenges faced in promoting tourism in Jharkhand. It provides a centralized and user-friendly system that enhances the overall travel experience.

By combining eco-tourism and cultural tourism, the platform not only attracts tourists but also promotes sustainability and cultural preservation. The use of modern technologies ensures scalability and performance.

In the future, the system can be improved by integrating AI-based recommendations, mobile applications, real-time booking systems, and multilingual support.

References :

- Jharkhand Tourism Official Website
- Spring Boot Documentation – <https://spring.io>
- Angular Documentation – <https://angular.io>
- MySQL Documentation – <https://www.mysql.com>
- GeeksforGeeks – Web Development Tutorials
- W3Schools – Web Technologies

